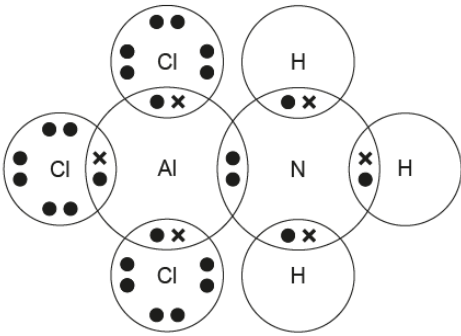


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				3d ⁷ allow any clear representation		1		1		
2				 NH₃ structure (1) dots and crosses showing coordinate bond and covalent bonds around Al atom (1)		2		2		
3				$K_w = [H^+][OH^-]$	1			1		
4				platinum as it is unreactive	1			1		
5				SiCl ₄ has available d-orbitals whilst carbon does not (so water can bond to Si atom)	1			1		
6				forms a gas and gases have a greater entropy than liquids (and solids)		1		1		